Name: ____________________

Class: ____________________

Date: ____________________

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 2 – Describing an Oscillation

Notes and Problem Set

Last lesson you measured an oscillator. This lesson we give the motion its proper language, and you find that the equation you tested experimentally is one of a family.

The defining equation

Key concept

$$a = -\omega^2 x$$

This is the definition of simple harmonic motion, not a result derived from something else. Read it as a sentence:

- acceleration is **proportional** to displacement — the further from equilibrium, the harder the object is pulled back;
- the constant of proportionality is ω^2, which is always positive;
- the **minus sign** makes a point opposite to x. Displace the object to the right and it accelerates to the left, and vice versa.

Remove the minus sign and the object accelerates *away* from equilibrium and never returns. The minus sign is the entire reason an oscillation oscillates.

Three ways to say how fast it repeats

Key concept

$$T = \frac{1}{f} = \frac{2\pi}{\omega}$$

- **Period** T — seconds for one full oscillation.
- **Frequency** f — oscillations per second, in Hz.
- **Angular frequency** ω — in rad/s. It is the one that appears in the equations, so it is worth being fluent in.

Angular frequency looks strange until you notice that one full oscillation is 2π radians of a repeating cycle. ω is simply how many radians of cycle pass per second.

Two oscillators worth knowing

Key concept

mass–spring: $T = 2\pi\sqrt{\dfrac{m}{k}}$ simple pendulum: $T = 2\pi\sqrt{\dfrac{l}{g}}$

You measured the first one. Look at what each period does *not* depend on:

- Neither depends on **amplitude** — which is what your amplitude check was testing.
- The pendulum does not depend on the **mass** of the bob at all. A heavy pendulum and a light one of the same length keep the same time.
- The pendulum *does* depend on g, which is why a pendulum clock is really a gravimeter, and why one taken to the Moon runs slow.

Energy over one cycle

Key concept

Over one oscillation, energy moves back and forth between kinetic and potential, and (with no friction) the total stays constant.

- **At maximum displacement** ($x = \pm x_0$): the object is momentarily at rest. Kinetic energy is zero; potential energy is at its maximum.
- **At the equilibrium position** ($x = 0$): the object is moving fastest. Kinetic energy is at its maximum; potential energy is at its minimum.
- **Everywhere between:** the two trade off, and their sum is the same everywhere.

Energy therefore completes **two** full cycles for every one cycle of the displacement — the object passes through the fast point twice per oscillation.

Higher level only

Phase angle, and solving oscillation problems algebraically.

Where the object is at $t = 0$ is set by the **phase angle** ϕ, measured in radians:

$$x = x_0 \sin(\omega t + \phi) \qquad v = \omega x_0 \cos(\omega t + \phi)$$

Two oscillators with the same ω but different ϕ are *out of phase*: they do the same thing at different moments.

Often you want speed at a *position* rather than at a time, and then the phase drops out entirely:

$$v = \pm\omega\sqrt{x_0^2 - x^2}$$

The $\pm$ is real physics: at a given displacement the object may be heading either way. Note that v is maximum, ωx_0, when $x = 0$, and zero when $x = \pm x_0$ — which is the energy story above, in algebra.

Quantitatively, the energies are

$$E_T = \tfrac{1}{2}m\omega^2 x_0^2 \qquad E_p = \tfrac{1}{2}m\omega^2 x^2$$

and kinetic energy is whatever is left: $E_k = E_T - E_p$.

Radians throughout. Every phase calculation in this topic is in radians. A calculator left in degrees is the single most common way to lose these marks.

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Problem set

Show your working. An answer with no working earns no method marks, and in this course the method is most of the mark.

1) **State** the two conditions that must hold for a motion to be simple harmonic.

[2]

2) **Explain** the significance of the minus sign in $a = -\omega^2 x$.

[2]

3) **Calculate** the period and the angular frequency of an object oscillating at $2.5\,\text{Hz}$.

[3]

4) **Determine** the period and frequency of a $0.150\,\text{kg}$ mass on a spring of spring constant $22\,\text{N/m}$.

[3]

5) **Calculate** the period of a simple pendulum of length $0.65\,\mathrm{m}$. Take $g = 9.81\,\mathrm{m/s^2}$.

[2]

6) **Determine** the length of a pendulum whose period is exactly $2.00\,\mathrm{s}$. (A pendulum like this beats once per second, and it is why grandfather clocks are the height they are.)

[3]

7) **Describe** how the kinetic and potential energy of an oscillator change during one complete cycle.

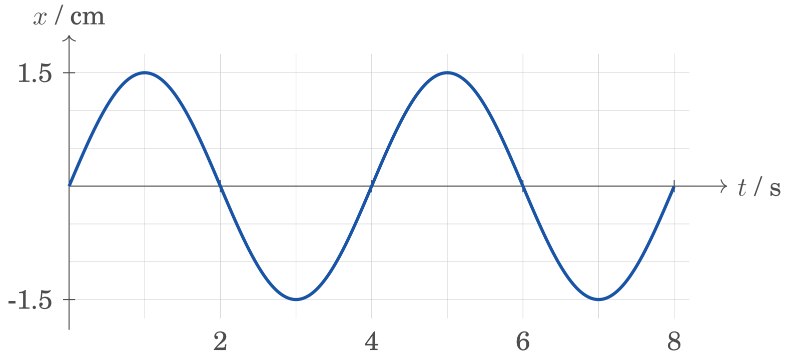
[3]

8) **Deduce** how the period changes if the mass on a spring is increased from m to $4m$, the spring being unchanged.

[2]

9) Determine the amplitude, period, frequency and angular frequency of the oscillation shown below.

[4]



Higher level

Higher level only

Questions 10–13 are higher level only. Work in **radians**.

10) Calculate the displacement at $t = 0.80\,\mathrm{s}$ of an oscillator described by $x = x_0 \sin(\omega t + \phi)$, where $x_0 = 4.0\,\mathrm{cm}$, $\omega = 3.0\,\mathrm{rad/s}$ and $\phi = 0$.

[3]

11) Determine the speed of that oscillator when its displacement is $2.0\,\mathrm{cm}$, and state its maximum speed.

[4]

12) **Calculate** the total energy of the oscillator in question 10 if its mass is $0.30\,\mathrm{kg}$.

[2]

13) **Determine** the potential and kinetic energies of that same oscillator at a displacement of $2.0\,\mathrm{cm}$, and check your kinetic energy against your answer to question 11.

[4]

Before next lesson

Your work for the fortnight

We do not meet again for two weeks, so this is a fortnight's work, not a night's. Spread it out — and start the write-up while the practical is still fresh, not the evening before.

1. **The reading** — set separately in ManageBac. Come to the next lesson having met the vocabulary.
2. **This problem set**, questions 1–9, and 10–13 if you are taking higher level.
3. **The full practical write-up** in your lab notebook, to the list you were given last lesson — including the honest evaluation.